Forensic Investigation Report

Case Subject: Incident Response: Suspicious System Intrusion • Case Number: **CASE-2026-001**

1. Investigation & Custody Metadata

Case Number:	CASE-2026-001	**Date of Report:**	2026-11-09 13:25:20
Lead Investigator:	Agent Vance (Chief Investigator)	**Case Status:**	Active
Organization:	Digital Forensics & Incident Response Lab	**Total Artifacts:**	22
Case Scope / Notes:	Primary digital evidence analysis case for triage and artifact verification.		

2. Executive Summary & Telemetry

This forensic examination involved the cryptographic acquisition, integrity verification, and AI-driven heuristic anomaly analysis of **22** digital evidence items totaling **0.08 MB**. Of the acquired evidence items, **22** were verified with intact cryptographic SHA-256 baselines. The AI Anomaly Detection engine flagged **8** items warranting prioritized investigator review, including **2** disguised file extension evasion attempts and **9** duplicate clusters.

22	22	8	9
EVIDENCE ITEMS	VERIFIED INTACT	ANOMALIES / RISKS	DUPLICATE SETS

3. Evidence Cryptographic Integrity Register (SHA-256)

File Name	Size	Detected Type	SHA-256 Cryptographic Hash	Status
wp-content-uploader.php	102 B	PHP Script	95c5442effea95f77d08117f80d9ab8785ecf4dc109871627ce5055762cb6f1b	Verified
server_rack_diagram.jpg	104 B	JPEG Image	a07771438b4a1427f54e00a8cd17655ac477a1974d12f209237eef928e4e7ea9	Verified
security_driver_mod.sys	1,026 B	Windows Executable (	93e621976de0815103ac545f13ecccdaa6e20a57a09bc86d7cd7837b5bf0c370	Verified
locked_database_payload.	32,768 B	Generic Binary / Unk	690296ee035bdfab7df25e318898f3d73f927c4b94145b0e4d8493292cfc4b22	Verified
invoice_urgent.pdf.exe	94 B	Windows Executable (	19191e62746e04515bd783a305dc5d040354e7b0a8a83d6abde7df3d56dfa168	Verified
incident_briefing.pdf	375 B	PDF Document	78506712cb7dd164465c2f68402dd214b1f25131b258680cc87f440c2ebdcfbe	Verified
HOW_TO_DECRYPT_FILES.txt	120 B	Plain Text Document	c7de9134f476939febf7342b437378fe8a32cca9e2883f9056eeb848624aa868	Verified

corporate_financials.jpg	616 B	Windows Executable (	163e39b11bf285d21096bec18fe4acd5f92b15da3818983e4190aaaeeb8fd831	Verified
backup_archive_secondary	1,560 B	ZIP Archive / OpenXM	49f3afe92f2b9f841beb717d41ad2a4f232e39fb99a9b7e0547fce1f9388c1b6	Verified
backup_archive_primary.z	1,560 B	ZIP Archive / OpenXM	49f3afe92f2b9f841beb717d41ad2a4f232e39fb99a9b7e0547fce1f9388c1b6	Verified
auth_system_access.log	3,588 B	Plain Text Document	646385ef73379ee39c838475be5d24211d5a99bb76bea975129aba3ac6f9aa90	Verified
wp-content-uploader.php	102 B	PHP Script	95c5442effea95f77d08117f80d9ab8785ecf4dc109871627ce5055762cb6f1b	Verified
server_rack_diagram.jpg	104 B	JPEG Image	a07771438b4a1427f54e00a8cd17655ac477a1974d12f209237eef928e4e7ea9	Verified
security_driver_mod.sys	1,026 B	Windows Executable (	93e621976de0815103ac545f13ecccdaa6e20a57a09bc86d7cd7837b5bf0c370	Verified
locked_database_payload.	32,768 B	Generic Binary / Unk	690296ee035bdfab7df25e318898f3d73f927c4b94145b0e4d8493292cfc4b22	Verified
invoice_urgent.pdf.exe	94 B	Windows Executable (	19191e62746e04515bd783a305dc5d040354e7b0a8a83d6abde7df3d56dfa168	Verified
incident_briefing.pdf	375 B	PDF Document	78506712cb7dd164465c2f68402dd214b1f25131b258680cc87f440c2ebdcfbe	Verified
HOW_TO_DECRYPT_FILES.txt	120 B	Plain Text Document	c7de9134f476939febf7342b437378fe8a32cca9e2883f9056eeb848624aa868	Verified
corporate_financials.jpg	616 B	Windows Executable (	163e39b11bf285d21096bec18fe4acd5f92b15da3818983e4190aaaeeb8fd831	Verified
backup_archive_secondary	1,560 B	ZIP Archive / OpenXM	49f3afe92f2b9f841beb717d41ad2a4f232e39fb99a9b7e0547fce1f9388c1b6	Verified
backup_archive_primary.z	1,560 B	ZIP Archive / OpenXM	49f3afe92f2b9f841beb717d41ad2a4f232e39fb99a9b7e0547fce1f9388c1b6	Verified
auth_system_access.log	3,588 B	Plain Text Document	de8b7e0027473eab39a8ccbe8c746bc76ce1daee00dffa7b8956e4f5946aeaaf	Verified

4. AI-Driven Anomaly & Threat Detection

File: security_driver_mod.sys • **Path:** D:\Digital forensics evidence analysis\sample_evidence\security_driver_mod.sys

SUSPICIOUS (Score: 41.2%)

Entropy: 2.02/8.0 • **MIME:** application/x-dosexec • **Detected:** Windows Executable (PE/DLL/EXE)

Forensic Findings:
• Modified time (2024-11-09T13:25:18.521858) precedes creation time (2026-11-09T13:15:05.532433) by 63071387 seconds.

File: locked_database_payload.bin • **Path:** D:\Digital forensics evidence analysis\sample_evidence\locked_database_payload.bin

CRITICAL ANOMALY (Score: 85.0%)

Entropy: 7.99/8.0 • **MIME:** application/octet-stream • **Detected:** Generic Binary / Unknown

Forensic Findings:
• High Shannon entropy (7.9947/8.0) strongly indicates encrypted or packed code.
• AI Alert: Filename contains known threat actor keyword match: 'locked_database_payload.bin'

File: invoice_urgent.pdf.exe • **Path:** D:\Digital forensics evidence analysis\sample_evidence\invoice_urgent.pdf.exe **SUSPICIOUS** (Score: 55.0%)

Entropy: 2.05/8.0 • **MIME:** application/x-dosexec • **Detected:** Windows Executable (PE/DLL/EXE)

Forensic Findings:
• Deceptive double extension detected: 'invoice_urgent.pdf.exe'

File: corporate_financials.jpg • **Path:** D:\Digital forensics evidence analysis\sample_evidence\corporate_financials.jpg **HIGH RISK** (Score: 65.0%)

Entropy: 1.36/8.0 • **MIME:** application/x-dosexec • **Detected:** Windows Executable (PE/DLL/EXE)

Forensic Findings:
• CRITICAL: Binary executable disguise! File has extension '.jpg' but magic bytes match 'Windows Executable (PE/DLL/EXE)'.
• AI Alert: Critical file type extension mismatch detected.

File: security_driver_mod.sys • **Path:** D:\Digital forensics evidence analysis\sample_evidence\security_driver_mod.sys **SUSPICIOUS** (Score: 41.2%)

Entropy: 2.02/8.0 • **MIME:** application/x-dosexec • **Detected:** Windows Executable (PE/DLL/EXE)

Forensic Findings:
• Modified time (2024-11-09T13:24:27.397390) precedes creation time (2026-11-09T13:15:05.532433) by 63071438 seconds.

File: locked_database_payload.bin • **Path:** D:\Digital forensics evidence analysis\sample_evidence\locked_database_payload.bin **CRITICAL ANOMALY** (Score: 85.0%)

Entropy: 7.99/8.0 • **MIME:** application/octet-stream • **Detected:** Generic Binary / Unknown

Forensic Findings:
• High Shannon entropy (7.9947/8.0) strongly indicates encrypted or packed code.
• AI Alert: Filename contains known threat actor keyword match: 'locked_database_payload.bin'

File: invoice_urgent.pdf.exe • **Path:** D:\Digital forensics evidence analysis\sample_evidence\invoice_urgent.pdf.exe **SUSPICIOUS** (Score: 55.0%)

Entropy: 2.05/8.0 • **MIME:** application/x-dosexec • **Detected:** Windows Executable (PE/DLL/EXE)

Forensic Findings:
• Deceptive double extension detected: 'invoice_urgent.pdf.exe'

File: corporate_financials.jpg • **Path:** D:\Digital forensics evidence analysis\sample_evidence\corporate_financials.jpg **HIGH RISK** (Score: 65.0%)

Entropy: 1.36/8.0 • **MIME:** application/x-dosexec • **Detected:** Windows Executable (PE/DLL/EXE)

Forensic Findings:
• CRITICAL: Binary executable disguise! File has extension '.jpg' but magic bytes match 'Windows Executable (PE/DLL/EXE)'.
• AI Alert: Critical file type extension mismatch detected.

5. Duplicate Evidence Identification

SHA-256 Hash	Copies	File Size	Wasted Volume	Associated File Paths
`49f3afe92f2b9f841beb...`	4	1,560 B	4,680 B	backup_archive_primary.zip backup_archive_secondary.zip backup_archive_primary.zip *+1 more*
`690296ee035bdfab7df2...`	2	32,768 B	32,768 B	locked_database_payload.bin locked_database_payload.bin
`93e621976de0815103ac...`	2	1,026 B	1,026 B	security_driver_mod.sys security_driver_mod.sys
`163e39b11bf285d21096...`	2	616 B	616 B	corporate_financials.jpg corporate_financials.jpg
`78506712cb7dd164465c...`	2	375 B	375 B	incident_briefing.pdf incident_briefing.pdf
`c7de9134f476939febf7...`	2	120 B	120 B	HOW_TO_DECRYPT_FILES.txt HOW_TO_DECRYPT_FILES.txt
`a07771438b4a1427f54e...`	2	104 B	104 B	server_rack_diagram.jpg server_rack_diagram.jpg
`95c5442effea95f77d08...`	2	102 B	102 B	wp-content-uploader.php wp-content-uploader.php
`19191e62746e04515bd7...`	2	94 B	94 B	invoice_urgent.pdf.exe invoice_urgent.pdf.exe

6. Cryptographic Chain of Custody & Audit Certification

All actions, evidence acquisitions, and algorithmic scans performed in this investigation are recorded in an immutable SHA-256 cryptographic hash-chained audit ledger.

Audit Chain Status: **INTEGRITY INTACT - CHAIN VALID**
Total Sequenced Records: 25
Head Cryptographic Proof: `8e636b05f2021a7c8da8974b00165c2c62c7855a399870a89b1244ffc0640fad`

7. Examiner Certification & Attestation

I hereby attest under official oath that the examination and analysis documented herein were performed in accordance with standard scientific digital forensic principles. The evidence integrity hashes were cryptographically computed and maintained without unauthorized alteration.

Examiner Signature: _____

Name: Agent Vance (Chief Investigator)
Title: Lead Forensic Investigator

Peer Reviewer: _____

Date: 2026-11-09
Agency: Digital Forensics & Incident Response Lab